CASE STUDY 1: WHY JAPAN RECYCLES MORE THAN MOST COUNTRIES: LESSONS FOR INDIA

Name: Kavish	Roll No / Div: 24102C0036	Subject: Technical Elective (TE) / MSE 2	Date: _____

1. EXECUTIVE SUMMARY

Japan built one of the world's most disciplined e-waste systems by legally tying manufacturers, retailers and consumers together. The Home Appliance Recycling Law (in force April 2001) makes producers recycle returned appliances to mandatory weight-based rates, and a 2013 law extended this to small devices. Results: high recovery rates, few illegal dumps, and a national 'urban mining' culture (Tokyo 2020 medals).

2. BACKGROUND

Japan has scarce natural resources and limited landfill space, so waste was treated as a resource problem. Law enacted in 1998, enforced April 2001, initially for 4 appliances: air conditioners, CRT TVs, refrigerators/freezers, washing machines. Reviewed after 5 years and expanded in 2009 (LCD/plasma TVs, clothes dryers).

3. STRATEGY / INITIATIVE

- Shared responsibility: consumer pays a recycling + collection fee when discarding; retailer collects; manufacturer must recycle at designated plants.
- Legally fixed minimum recycling rates (by weight); targets raised in 2009 (e.g. AC 60→70%, fridge 50→60%, washing machine 50→65%).
- Recycling tag/manifest system lets consumers trace their appliance online; fluorocarbons recovered and destroyed.
- Small Home Appliance Recycling Act (2013) + Tokyo 2020 Medal Project: medals made from donated phones and small electronics.

4. IMPACT & RESULTS (DATA TABLE)

Indicator	Value
Appliances collected FY2003 / FY2005	10.46 million / 11.62 million units
Recycling rate FY2005 (target)	AC 84% (60), TV 77% (55), Fridge 66% (50), W. machine 75% (50)
Recycling plants (2004)	41 plants, ~2,200 jobs
2009 change	LCD/plasma TVs + dryers added; targets raised

5. SWOT ANALYSIS

Strengths: Clear legal duties for all; fee-based funding; traceability; tech-heavy dismantling	**Weaknesses:** Consumer pays at disposal (fly-tipping risk); law covers limited items; targets are weight-based, not toxicity-based
Opportunities: Urban mining of gold/silver/copper; design-for-environment; export of the model to Asia	**Threats:** Rising volumes of small/IoT devices; illegal export of used appliances

6. COMPARISON WITH INDIA

India (2023-24): 1.75 Mt e-waste generated, 0.76 Mt recycled (~43%); EPR under E-Waste Rules 2022 covers 106 items but informal sector still dominates. Japan's rates measure recovery of appliances that reach recyclers, so they are not directly comparable. Key difference: Japan has strong retailer take-back and a visible end-of-life fee; India relies on producer EPR certificates and has weak consumer-facing collection.

7. RECOMMENDATIONS

(1) Mandatory retailer take-back with clear disposal points; (2) trackable tag/QR system for each large appliance; (3) raise recycling targets stepwise with periodic law review; (4) run a phone/small-device drive like the Medal Project to build public awareness.

8. CONCLUSION

Japan shows that recycling succeeds when law, money flow and public habit are aligned. India has the EPR framework; it now needs Japan-style collection convenience, traceability and enforcement.

9. REFERENCES

MoE Japan / Hosoda (Keio Univ.), 'Japan's Home Appliance Recycling Law' (2006); Japan for Sustainability news (2004, 2006, 2009); Hitachi Hyoron (2020); Rajya Sabha reply on e-waste (Aug 2024).

10. ONE-PAGE EXAM REVISION SHEET

- Law: enacted 1998, in force Apr 2001; 4 items → 6 items in 2009; small appliance act 2013.
- Model: consumer fee + retailer collects + maker recycles (designated plants).
- FY2005 rates beat targets: AC 84, TV 77, fridge 66, WM 75 (%).
- Tokyo 2020 medals = urban mining of small electronics.
- India gap: 43% formal recycling vs need for retailer take-back + traceability.
- Viva: Why did Japan review the law after 5 years? How is it different from India's EPR?

CASE STUDY 2: SWITZERLAND'S E-WASTE SUCCESS STORY

Name: Kavish	Roll No / Div: 24102C0036	Subject: Technical Elective (TE) / MSE 2	Date: _____

1. EXECUTIVE SUMMARY

Switzerland runs one of the oldest and best-performing take-back systems in the world. Consumers pay an Advance Recycling Fee (ARF) inside the purchase price, so returning old devices is free. SENS eRecycling alone collected 104,751 tonnes (11.48 kg per resident) in 2025.

2. BACKGROUND

SENS eRecycling was founded in 1990, well before most countries had e-waste rules; it is a foundation set up by the industry. Legal basis is the Swiss ordinance on returning, taking back and disposing of electrical equipment (ORDEE/VREG). A second system, Swico, handles IT, telecom and consumer electronics. Together they collected ~15.3 kg per person (2021).

3. STRATEGY / INITIATIVE

- Advance Recycling Fee (ARF) paid by producers/importers, added to price; funds collection, transport and treatment.
- Free return at any retailer (even without buying new) or at 634 official collection points; ~1,557 producers/importers subject to ARF.
- Only licensed recyclers (17 in 2025, all in Switzerland); yearly audits of storage, treatment and reporting.
- Export of defective appliances needs a federal permit, which limits dumping abroad.

4. IMPACT & RESULTS (DATA TABLE)

Indicator	Value
SENS collection 2025	104,751 t = 11.48 kg per resident
Total (SENS+Swico) 2021	~15.3 kg per person
Since 1990 (SENS)	over 1.4 million tonnes collected
Network 2025	634 collection points, 17 recyclers
Material recovery	~75% of an appliance reused (plastics, metals, glass)

5. SWOT ANALYSIS

Strengths: Prepaid ARF = free, convenient return; dense network; producer-run and audited; local recycling	**Weaknesses:** High costs; small-country model; system depends on public discipline
Opportunities: Cover new streams: solar panels, batteries, e-cigarettes, heat pumps (already added)	**Threats:** Fast-growing battery/PV waste; online sellers avoiding fees; critical-metal price swings

6. COMPARISON WITH INDIA

India generated ~1.75 Mt in 2023-24 (about 1.2 kg per person, derived from population ~1.4 billion) versus Switzerland's ~15 kg collected per person. India's EPR uses producer targets and tradable certificates, but consumers receive no easy free-return channel and most flow goes to informal dealers. Switzerland's ARF is visible and universal; India has no equivalent prepaid fee.

7. RECOMMENDATIONS

(1) Pilot a prepaid disposal levy on new electronics in India; (2) mandatory retailer/brand take-back points in every district; (3) audit and publish recycler data; (4) restrict informal exports/scrap trading via licensing.

8. CONCLUSION

Switzerland proves a producer-funded prepaid model gives high collection with no consumer friction. India can adapt the ARF idea within EPR to move e-waste from informal to formal channels.

9. REFERENCES

SENS eRecycling: facts & figures 2025, 'Making eRecycling a success'; SENS factsheet (WEEE Forum); Umweltperspektiven.ch (2017); Rajya Sabha PQ 265 (Aug 2024).

10. ONE-PAGE EXAM REVISION SHEET

- SENS founded 1990; financed by ARF; Swico handles IT/telecom.
- 2025: 104,751 t, 11.48 kg per person; 634 points, 17 recyclers.
- Free return at any shop, no purchase needed.
- ~75% of material reusable; export needs permit.
- India: no prepaid fee, informal dominance, ~1.2 kg per person generated.
- Viva: How does ARF work? Why is Swiss collection per capita so high?

CASE STUDY 7: KERALA'S CLEAN KERALA COMPANY (CKCL)

Name: kavish	Roll No / Div: 24102c0036	Subject: Technical Elective (TE) / MSE 2	Date: _____

1. EXECUTIVE SUMMARY

Clean Kerala Company Ltd, a Government of Kerala company, buys segregated non-biodegradable waste from Haritha Karma Sena (HKS) units run by Kudumbashree women and sends it for recycling or scientific disposal. Collection grew from 7,657 t (2021-22) to 86,295 t (2025-26). In 2025 the model added e-waste collection.

2. BACKGROUND

Kerala follows Solid Waste Management Rules 2016, which made the Green Task Force (HKS) responsible for door-to-door collection of inorganic waste in every local body. Kerala generates ~3.7 million tonnes of solid waste yearly (CAG 2022-23), so waste was a livelihood and an environmental issue.

3. STRATEGY / INITIATIVE

- Decentralised model: HKS collects from homes/shops and charges a monthly user fee.
- Waste goes via MCFs/MRFs to CKCL, which pays HKS for segregated plastics and other recyclables.
- Aug 2025 e-waste drive: HKS buys 44 types of non-hazardous e-items by weight from homes/institutions; CKCL hands them to authorised recyclers.
- Livelihood spin-offs: 223 cloth-bag and 540 paper-bag units; 400+ micro-enterprises.

4. IMPACT & RESULTS (DATA TABLE)

Year / item	Result
2021-22 waste received by CKCL	7,657 tonnes
2022-23	30,217 tonnes (about 4×)
2024-25	50,190 t; 35,214 women; 4,438 HKS units; user fees Rs 341 crore
2025-26	86,295.7 t (highest ever); 37,481 women; Rs 11.49 crore from waste sales
E-waste drive (1st month, 2025)	33,945 kg collected; Rs 2.63 lakh paid out

5. SWOT ANALYSIS

Strengths: Women-led livelihood; strong local-body network; steady CKCL buyer; scale across the state	**Weaknesses:** E-waste only non-hazardous items; low pay per kg; depends on user fees and local body funds
Opportunities: Link with EPR producers/PROs to get recycler payments; expand e-waste and tablet-strip recycling	**Threats:** Logistics costs of transporting low-value plastic; informal dealers; hazardous e-waste not covered

6. COMPARISON WITH INDIA

India's national formal recycling is ~43% of generated e-waste (2023-24) with the informal sector dominating. Kerala shows local-body-led collection with a fixed buyer can formalise waste flows. Kerala's waste moves through a defined chain (HKS to CKCL). Gap: Kerala's numbers are mostly plastic/dry waste, and e-waste volume is still small.

7. RECOMMENDATIONS

(1) Extend HKS e-waste collection to all districts with training on safe handling; (2) sign EPR tie-ups with producers so HKS gets paid per tonne; (3) install digital weight and payment tracking (Harithamitram app); (4) set up drop boxes at colleges and offices.

8. CONCLUSION

CKCL demonstrates that a decentralised, women-led collection network with a guaranteed buyer can rapidly scale waste formalisation. Its e-waste extension needs EPR funding and hazardous-waste handling to match its plastic success.

9. REFERENCES

LSGD Kerala/Kudumbashree (Mar 2025); UNI India (Aug 2026); Onmanorama (Jun 2023); Kerala Kaumudi (Aug 2025); ISB Bharti Institute blog on HKS model.

10. ONE-PAGE EXAM REVISION SHEET

- CKCL = state company buying waste from HKS (Kudumbashree women).
- Growth: 7,657 t (2021-22) → 30,217 t → 50,190 t → 86,295 t (2025-26).
- E-waste drive Aug 2025: 44 items, 33,945 kg in a month.
- Model = decentralised + user fee + guaranteed buyer.
- Gap vs India: needs EPR link and hazardous e-waste capability.
- Viva: What is HKS? How is CKCL funded and who pays HKS?

CASE STUDY 5: BENGALURU: INDIA'S E-WASTE CAPITAL?

Name: kavish	Roll No / Div: 24102C0036	Subject: Technical Elective (TE) / MSE 2	Date: _____

1. EXECUTIVE SUMMARY

Bengaluru, India's IT capital, is among the largest e-waste generating cities, yet only a small share reaches authorised recyclers. Around 92,000 t/year was estimated in 2016 (third after Mumbai and Delhi), while the KSPCB says the formal sector handles only 10-15%. The city is a hub of both e-waste generation and informal recycling.

2. BACKGROUND

Home to thousands of software firms, data centres and a large consumer base, Bengaluru discards computers, phones and appliances at high rates. Assocham-KPMG (2016) ranked it third in India. KSPCB reports ~1,000 tonnes/month from IT firms alone. Informal hubs such as Goripalya, Shivajinagar and KG Halli extract metals using acids and open burning.

3. STRATEGY / INITIATIVE

- KSPCB authorises collection centres, dismantlers and recyclers (27 collection centres, 42 dismantlers, 24 recyclers, 14 recycler-cum-dismantlers at one point) under E-Waste Rules 2016.
- EPR and bulk-consumer rules: IT firms must channel waste to authorised recyclers.
- Awareness drives by KSPCB and PROs like Saahas Zero Waste; startups (e.g. BinBag) offer doorstep pickup.
- Centre announced India's first government-run e-waste recycling plant in Bengaluru (2018).

4. IMPACT & RESULTS (DATA TABLE)

Indicator	Value
Annual e-waste (2016 Assocham-KPMG)	~92,000 tonnes; 3rd after Mumbai, Delhi
IT-sector e-waste (KSPCB)	~1,000 tonnes per month
Formal sector share	only 10-15% (KSPCB, after 10 years of rules)
Informal practitioners	~4,000 estimated (Deccan Herald)
Household e-waste	~25 kg per middle-income home/yr (BinBag est.)

5. SWOT ANALYSIS

Strengths: Strong IT base; many authorised units; active startups and NGOs; KSPCB enforcement drives	**Weaknesses:** Informal sector dominance; poor consumer awareness; no city-level data on generation
Opportunities: Corporate EPR and bulk-consumer compliance; Data-tracked pickup apps; skill training for informal workers	**Threats:** Toxic exposure of workers; soil/water pollution; data leakage from unwiped devices

6. COMPARISON WITH INDIA

Nationally, India recycled ~43% of e-waste formally in 2023-24 and reported ~70.7% recovery in 2024-25 under EPR portal data. Bengaluru's formal share of 10-15% (older KSPCB statement, so figures are not the same year) is far below the national average. Volume alone does not make Bengaluru the capital: Mumbai and Delhi generated more in 2016, but Bengaluru is the IT e-waste capital.

7. RECOMMENDATIONS

(1) Compulsory IT-asset disposal audit for large companies; (2) integrate informal dealers into licensed collection with pricing incentives; (3) city-wide take-back kiosks and app-based pickup; (4) publish yearly city e-waste data.

8. CONCLUSION

Bengaluru is India's IT e-waste capital in terms of source, but not a model of formal recycling. Bridging the informal gap through incentives, audits and data will decide whether it becomes a leader.

9. REFERENCES

Deccan Herald (KSPCB e-waste reports); Citizen Matters (Assocham data); The News Minute / TOI (2018); CPCB data via Rajya Sabha (2024); Drishti IAS (2025).

10. ONE-PAGE EXAM REVISION SHEET

- Bengaluru ≈ 92,000 t/yr (2016), 3rd after Mumbai & Delhi.
- IT firms: ~1,000 t/month; formal share 10-15% (KSPCB).
- Informal hubs: Goripalya, Shivajinagar, KG Halli.
- Tools: KSPCB authorisation, EPR, bulk consumer rules.
- Title debate: IT e-waste capital yes, largest total no.
- Viva: Why does informal recycling persist? What are the health risks?

CASE STUDY 10: DELL'S CIRCULAR ECONOMY STRATEGY

Name: Kavish	Roll No / Div: 24102C0036	Subject: Technical Elective (TE) / MSE 2	Date: _____

1. EXECUTIVE SUMMARY

Dell combines free take-back, refurbishing and closed-loop recycling to reuse materials in new devices. It reached its 100 million pounds recycled-content goal ahead of 2020 and now targets a 2030 'moonshot': recycle or reuse an equivalent product for every product sold, more than half of product content recycled or renewable, and 100% recycled/renewable packaging.

2. BACKGROUND

Dell has offered recycling for 20+ years and runs services in 80+ countries. Dell Reconnect (with Goodwill, ~2,000 US drop-off sites) started the consumer take-back base. In 2014 Dell launched closed-loop plastics with Wistron GreenTech, turning ABS plastic from returned computers into new parts.

3. STRATEGY / INITIATIVE

- Take-back and reuse: consumer and business recycling, asset resale, refurbishing before recycling.
- Closed-loop plastics blended at ~35% recycled content with virgin plastic; some products up to 50% recycled content.
- Reclaimed aerospace carbon fiber in Latitude laptops; bioplastic and recycled materials in lids (71% recycled/renewable in one case).
- Packaging: bamboo, recycled pulp, ocean-bound plastic; 100% recycled/renewable commercial laptop packaging.

4. IMPACT & RESULTS (DATA TABLE)

Indicator	Value
Recycled content goal	100 M lb (2020) achieved early
Closed-loop plastic parts	100M+ lb used on 125+ product lines (2021)
Used electronics recovered FY2017	177 M lb; 1.8 bn lb cumulative since FY2008
Carbon footprint	~11% lower vs virgin plastic (closed-loop)
Carbon fibre diverted	2M+ lb from landfill

5. SWOT ANALYSIS

Strengths: Global take-back reach; design for reuse; scale in supply chain; public targets	**Weaknesses:** Recycled plastic must be blended with virgin; only part of product line uses recycled content; Dell reports own data
Opportunities: Refurbished device market; recovered metals and rare earths; partnerships (Circular Electronics Partnership)	**Threats:** Greenwashing criticism; cost competition; different rules across countries

6. COMPARISON WITH INDIA

In India, producers must meet EPR recycling targets by buying certificates from registered recyclers, which focuses on end-of-life compliance rather than closed-loop material reuse. Dell also offers take-back and recycling services in India, but Indian rules do not yet require recycled-content or design-for-recycling standards for products.

7. RECOMMENDATIONS

(1) Indian OEMs to publish recycled-content targets for products; (2) offer trade-in and refurbish schemes with PROs; (3) develop domestic recycled plastic and metal supply chains for electronics; (4) BIS-style standards for recycled content and design-for-disassembly.

8. CONCLUSION

Dell shows that a producer-led circular strategy can turn waste into feedstock and cut emissions. For India, similar strategies would move EPR from paper compliance to real material recovery.

9. REFERENCES

Dell Technologies blog 'The Road to 2030 Must Be Circular'; Waste360 (2014, 2017); Business Today (May 2021); Sustainable Brands; Dell sustainable devices page.

10. ONE-PAGE EXAM REVISION SHEET

- Dell Reconnect + closed-loop plastics (2014) + 2030 goals (3 targets).
- Closed-loop blend ~35% recycled; 100M+ lb parts; 125+ lines.
- FY2017 recovered 177 M lb; 1.8 bn lb since FY2008.
- Carbon fibre reclaimed from aerospace.
- India: EPR certificates vs Dell's closed-loop material reuse.
- Viva: Define circular economy; how does closed-loop differ from recycling?